Mar 31 16:49

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate | vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

```
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine start alpha ./rootfs-alpha "/bin/sh -c 'echo alpha alive; sleep 1000'"
Container alpha already running
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine start beta ./rootfs-beta "/bin/sh -c 'echo beta alive; sleep 1000'"
Container beta already running
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
Expected states include: starting, running, stopped, killed, exited
beta    10843   running 40      64      Tue Mar 31 16:47:02 2026

alpha   10837   running 40      64      Tue Mar 31 16:47:02 2026

vk12345@ubuntu:~/OS-Jackfruit/boilerplate$
```

Mar 31 16:50

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate | vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

```
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine logs alpha
alpha alive
alpha alive
alpha alive
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine logs beta
beta alive
beta alive
beta alive
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$
```

Mar 31 16:53

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate | vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

```
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine start mem3 ./rootfs-mem "/bin/sh -c '/memory_hog 4 500'" --soft-mib 16 --hard-m
ib 32
Started container mem3
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sleep 4
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
Expected states include: starting, running, stopped, killed, exited
mem3    10895   killed  16      32      Tue Mar 31 16:51:57 2026

beta    10843   running 40      64      Tue Mar 31 16:47:02 2026

alpha   10837   running 40      64      Tue Mar 31 16:47:02 2026

vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine logs mem3
allocation=1 chunk=4MB total=4MB
allocation=2 chunk=4MB total=8MB
allocation=3 chunk=4MB total=12MB
allocation=4 chunk=4MB total=16MB
allocation=5 chunk=4MB total=20MB
allocation=6 chunk=4MB total=24MB
allocation=7 chunk=4MB total=28MB
allocation=8 chunk=4MB total=32MB
allocation=1 chunk=4MB total=4MB
allocation=2 chunk=4MB total=8MB
allocation=3 chunk=4MB total=12MB
allocation=4 chunk=4MB total=16MB
allocation=5 chunk=4MB total=20MB
allocation=6 chunk=4MB total=24MB
allocation=7 chunk=4MB total=28MB
allocation=8 chunk=4MB total=32MB
allocation=1 chunk=4MB total=4MB
allocation=2 chunk=4MB total=8MB
allocation=3 chunk=4MB total=12MB
allocation=4 chunk=4MB total=16MB
allocation=5 chunk=4MB total=20MB
allocation=6 chunk=4MB total=24MB
```

Mar 31 16:53

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate | vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

```
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
Expected states include: starting, running, stopped, killed, exited
mem3    10895   killed  16      32      Tue Mar 31 16:51:57 2026

beta    10843   running 40      64      Tue Mar 31 16:47:02 2026

alpha   10837   running 40      64      Tue Mar 31 16:47:02 2026

vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine logs mem3
allocation=1 chunk=4MB total=4MB
allocation=2 chunk=4MB total=8MB
allocation=3 chunk=4MB total=12MB
allocation=4 chunk=4MB total=16MB
allocation=5 chunk=4MB total=20MB
allocation=6 chunk=4MB total=24MB
allocation=7 chunk=4MB total=28MB
allocation=8 chunk=4MB total=32MB
allocation=1 chunk=4MB total=4MB
allocation=2 chunk=4MB total=8MB
allocation=3 chunk=4MB total=12MB
allocation=4 chunk=4MB total=16MB
allocation=5 chunk=4MB total=20MB
allocation=6 chunk=4MB total=24MB
allocation=7 chunk=4MB total=28MB
allocation=8 chunk=4MB total=32MB
allocation=1 chunk=4MB total=4MB
allocation=2 chunk=4MB total=8MB
allocation=3 chunk=4MB total=12MB
allocation=4 chunk=4MB total=16MB
allocation=5 chunk=4MB total=20MB
allocation=6 chunk=4MB total=24MB
allocation=7 chunk=4MB total=28MB
allocation=8 chunk=4MB total=32MB
allocation=9 chunk=4MB total=36MB
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$
```

Mar 31 16:53

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate | vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

```
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo dmesg | tail -n 20
[ 9110.607644] [container_monitor] Registering container=beta pid=9892 soft=41943040 hard=67108864
[ 9876.802352] [container_monitor] Registering container=mem3 pid=10071 soft=16777216 hard=33554432
[ 9878.461274] [container_monitor] SOFT LIMIT container=mem3 pid=10071 rss=17170432 limit=16777216
[ 9880.506934] [container_monitor] HARD LIMIT container=mem3 pid=10071 rss=33947648 limit=33554432
[ 9880.515552] [container_monitor] Unregister request container=mem3 pid=10071
[ 9968.954566] [container_monitor] Registering container=cpu1 pid=10088 soft=41943040 hard=67108864
[ 9988.593018] [container_monitor] Unregister request container=cpu1 pid=10088
[10110.105711] [container_monitor] Unregister request container=alpha pid=9886
[10110.115265] [container_monitor] Unregister request container=beta pid=9892
[10670.452249] [container_monitor] Module unloaded.
[11019.767306] hrtimer: interrupt took 26362175 ns
[11019.964969] kauditd_printk_skb: 135 callbacks suppressed
[11019.964972] audit: type=1400 audit(1774974591.341:146): apparmor="DENIED" operation="capable" class="cap" profile="/usr/sbin/cupsd" p
id=10379 comm="cupsd" capability=12  capname="net_admin"
[11664.339417] [container_monitor] Module loaded. Device: /dev/container_monitor
[12051.042912] [container_monitor] Registering container=alpha pid=10837 soft=41943040 hard=67108864
[12051.058796] [container_monitor] Registering container=beta pid=10843 soft=41943040 hard=67108864
[12346.478415] [container_monitor] Registering container=mem3 pid=10895 soft=16777216 hard=33554432
[12348.525257] [container_monitor] SOFT LIMIT container=mem3 pid=10895 rss=17170432 limit=16777216
[12350.578265] [container_monitor] HARD LIMIT container=mem3 pid=10895 rss=38141952 limit=33554432
[12350.588228] [container_monitor] Unregister request container=mem3 pid=10895
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$
```

Mar 31 16:55

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate | vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

```
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine start cpu1 ./rootfs-cpu "/bin/sh -c '/cpu_hog 20'"
Started container cpu1
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine start io1 ./rootfs-io "/bin/sh -c '/io_pulse 20 100'"
Started container io1
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sleep 3
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
Expected states include: starting, running, stopped, killed, exited
io1     10933   exited  40      64      Tue Mar 31 16:54:50 2026

cpu1    10927   exited  40      64      Tue Mar 31 16:54:39 2026

mem3    10895   killed  16      32      Tue Mar 31 16:51:57 2026

beta    10843   running 40      64      Tue Mar 31 16:47:02 2026

alpha   10837   running 40      64      Tue Mar 31 16:47:02 2026

vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine logs cpu1
cpu_hog alive elapsed=1 accumulator=6003449791739410138
cpu_hog alive elapsed=2 accumulator=11100206258616821430
cpu_hog alive elapsed=3 accumulator=12596938291846865950
cpu_hog alive elapsed=4 accumulator=14413869018238071917
cpu_hog alive elapsed=5 accumulator=6653871525390712862
cpu_hog alive elapsed=6 accumulator=11942479821650139819
cpu_hog alive elapsed=7 accumulator=13880500382421305226
cpu_hog alive elapsed=8 accumulator=6609215629228569197
cpu_hog alive elapsed=9 accumulator=6374240787686574835
cpu_hog alive elapsed=10 accumulator=15546144593061269220
cpu_hog alive elapsed=11 accumulator=2617912847337343779
cpu_hog alive elapsed=12 accumulator=9604790193788469533
cpu_hog alive elapsed=13 accumulator=14951924830131501011
cpu_hog alive elapsed=14 accumulator=15119940131611009230
cpu_hog alive elapsed=15 accumulator=3470564937798520856
cpu_hog alive elapsed=16 accumulator=12858545698405274126
cpu_hog alive elapsed=17 accumulator=4659631927581376961
```

```
cpu_hog done duration=20 accumulator=16435558465215250253
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine logs io1
io_pulse wrote iteration=1
io_pulse wrote iteration=2
io_pulse wrote iteration=3
io_pulse wrote iteration=4
io_pulse wrote iteration=5
io_pulse wrote iteration=6
io_pulse wrote iteration=7
io_pulse wrote iteration=8
io_pulse wrote iteration=9
io_pulse wrote iteration=10
io_pulse wrote iteration=11
io_pulse wrote iteration=12
io_pulse wrote iteration=13
io_pulse wrote iteration=14
io_pulse wrote iteration=15
io_pulse wrote iteration=16
io_pulse wrote iteration=17
io_pulse wrote iteration=18
io_pulse wrote iteration=19
io_pulse wrote iteration=20
io_pulse wrote iteration=1
io_pulse wrote iteration=2
io_pulse wrote iteration=3
io_pulse wrote iteration=4
io_pulse wrote iteration=5
io_pulse wrote iteration=6
io_pulse wrote iteration=7
io_pulse wrote iteration=8
io_pulse wrote iteration=9
io_pulse wrote iteration=10
io_pulse wrote iteration=11
io_pulse wrote iteration=12
io_pulse wrote iteration=13
```

Mar 31 16:57

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate | vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

```
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine stop alpha
Sent stop signal to alpha
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$
```

Mar 31 19:06

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

vk12345@ubuntu: ~/OS-Jackfruit/boilerplate | vk12345@ubuntu: ~/OS-Jackfruit/boilerplate

```
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ ps aux | grep defunct
vk12345    13397  0.0  0.0   6140  2100 pts/2    S+   19:06   0:00 grep --color=auto defunct
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$ lsmod | grep monitor
vk12345@ubuntu:~/OS-Jackfruit/boilerplate$
```